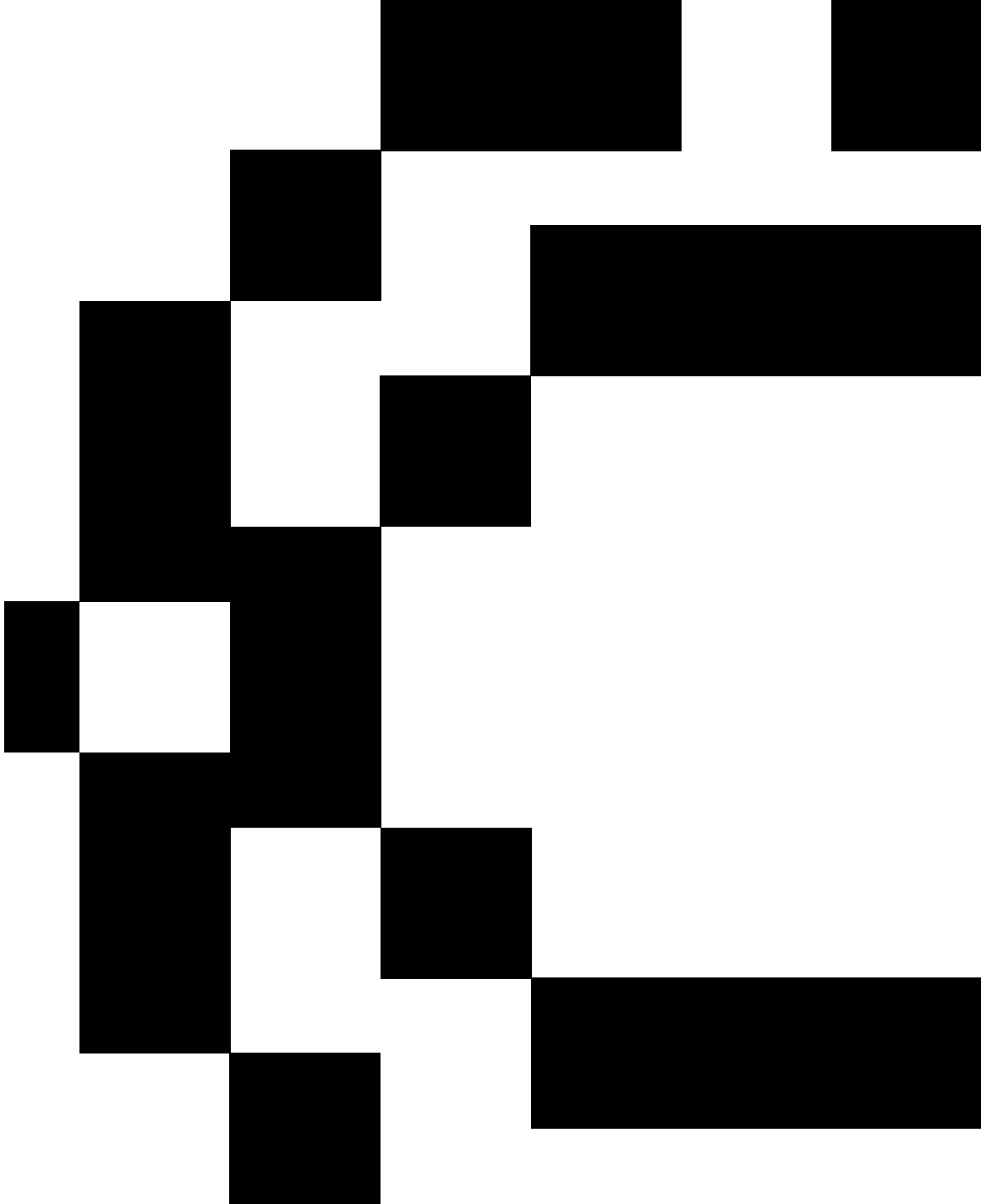
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

Practicals Week 09 - Exercises I

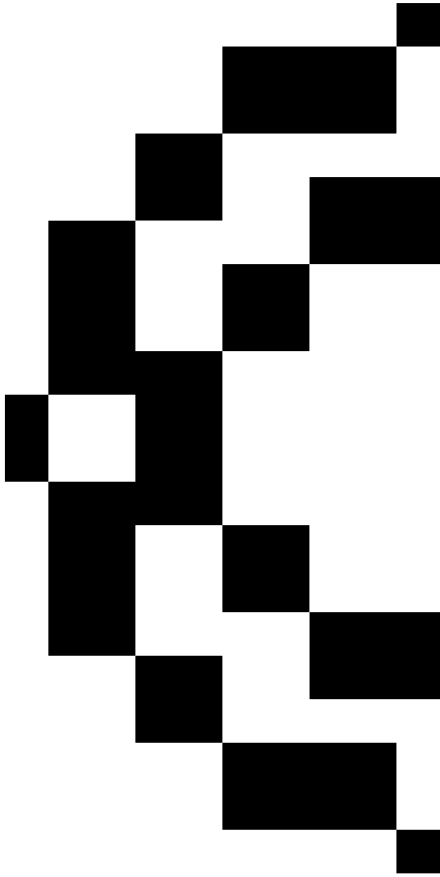


Exercise 1

Define two lists of positive integers of your choosing. Print the numbers that are common to both lists.

Example

- Input: `l1 = [1, 2, 3, 4, 5], l2 = [5, 8, 2, 9]`
- Output: `2`
- `5`

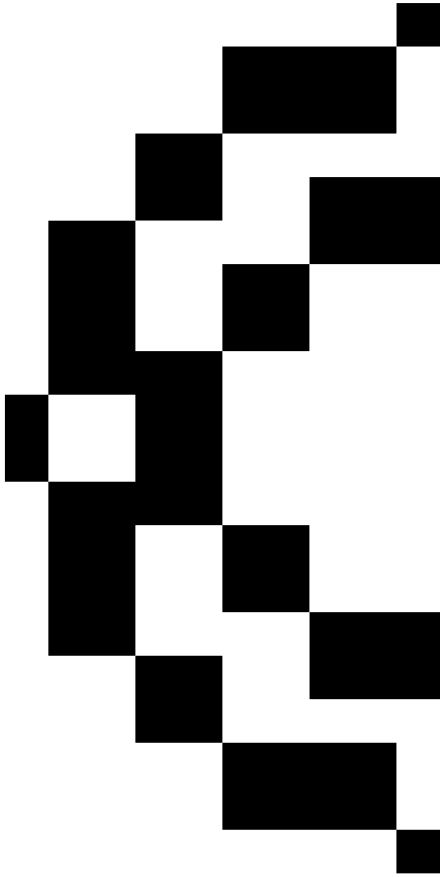


Exercise 2

Ask the user for 5 integers, one at a time, and store them in a list. Then, print every number in the list that is greater than 10, together with its index.

Example

```
Input:  numbers = 3, 15, 8, 22, 5
Output: 15 is bigger than 10 and at index 1
        22 is bigger than 10 and at index 3
```

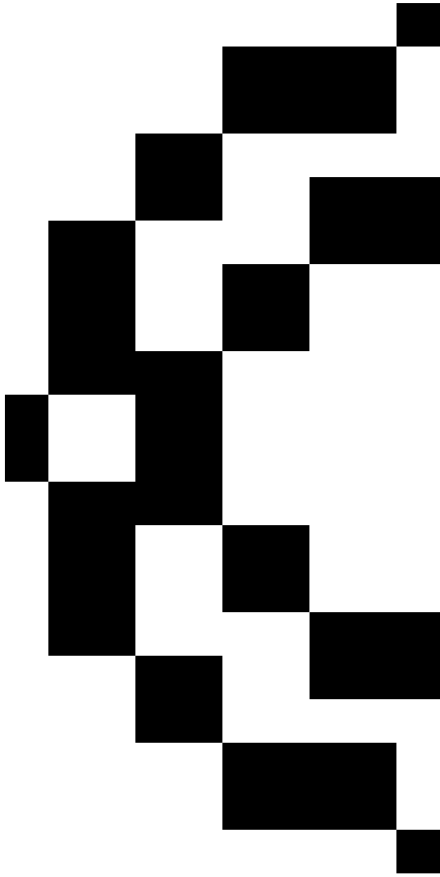


Exercise 3

Ask the user how many friends they would like to add to an invitation list. Ask for each friend's name and store the names in a list. Print how many of the names start with the letter A.

Example

- `Input:  friends = "Ana", "Bruno", "Alex"`
- `Output: 2 people's name starts with an A.`

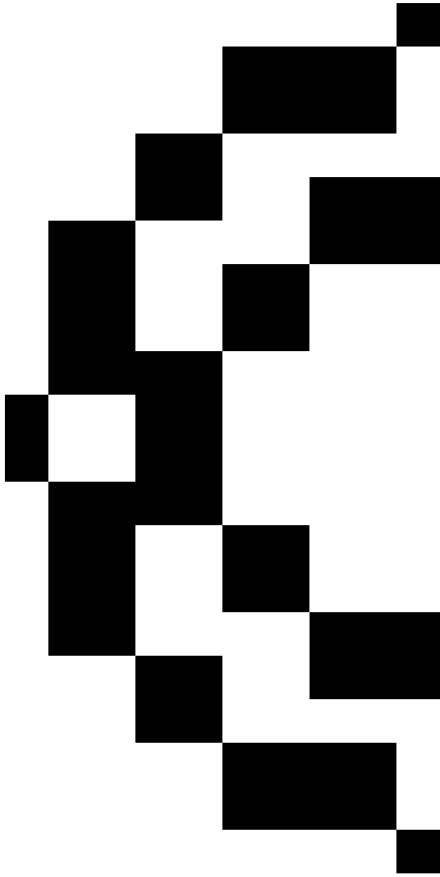


Exercise 4

Consider the list
`names = ["Anthony", "Gertrude", "Gertrudo", "Anthonella"]`. Go through all the names and, each time a vowel is found, print `"Found one!"`. For the list above, this should be printed 12 times in total.

Example

- `Input: names = ["Anthony", "Gertrude", "Gertrudo", "Anthonella"]`
- `Output: Found one!`
- `Found one!`
- `... (12 times in total)`

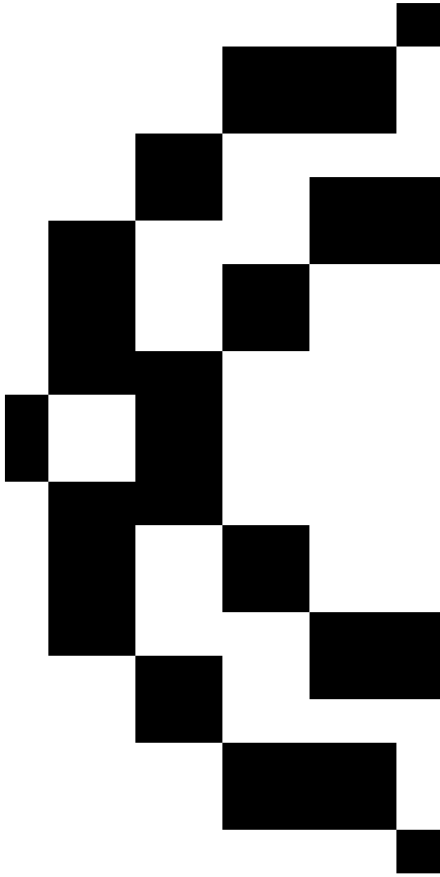


Exercise 5

Using nested `for` loops, print a grid of 5 rows and 3 columns of the `*` symbol.

Example

```
Input:  (no input)
Output: * * *
        * * *
        * * *
        * * *
        * * *
```

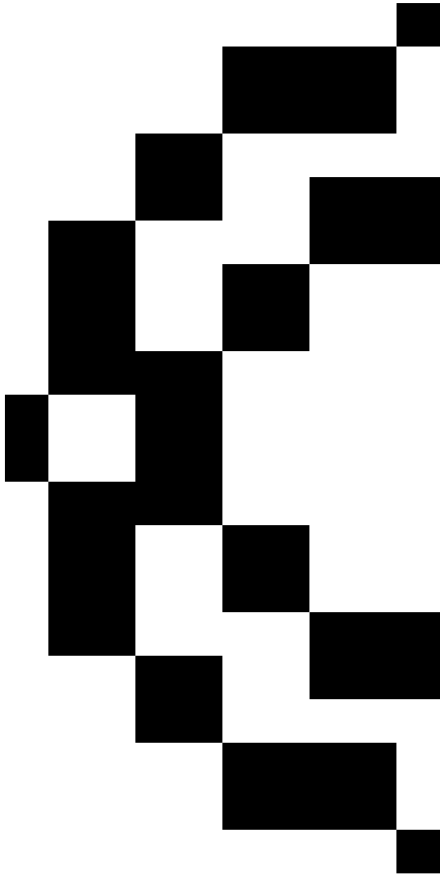


Exercise 6

Ask the user what symbol they would like to print, then ask how many rows and how many columns they would like. Print a grid of that symbol with the requested dimensions.

Example

```
Input:  symbol = "#", rows = 3, columns = 2
Output: # #
        # #
        # #
```

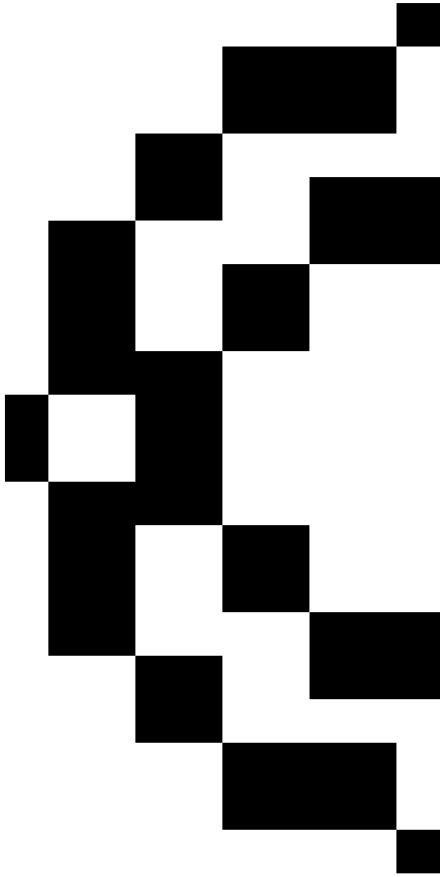


Exercise 7

Ask the user to insert their final grade. The program should not proceed until the grade entered is between 0 and 20 (inclusive). Once a valid grade is given, congratulate the user if it is above 17, otherwise print `":)"`. You may assume the user always types an integer.

Example

- `Input:  grade = 25, then 14`
- `Output: :)`

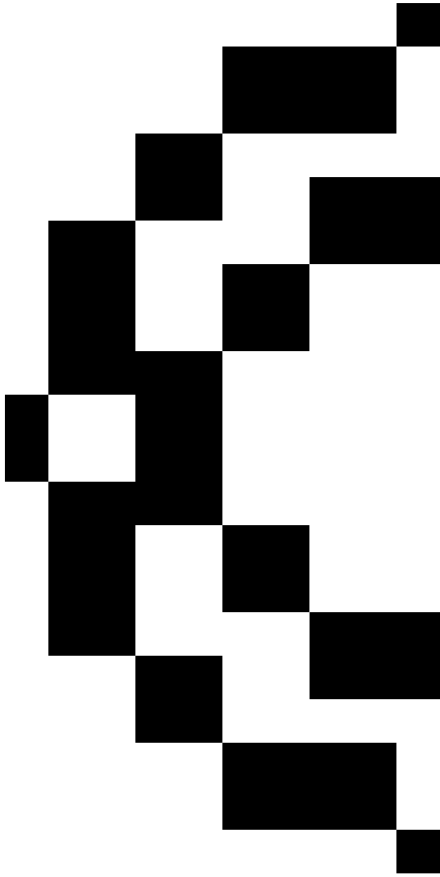


Exercise 8

Repeat the previous exercise, but this time do not assume the user types an integer — the program must keep asking until it receives text that is both a whole number and within the range 0 to 20.

Example

- `Input:  grade = "abc", then "25", then "14"`
- `Output: :)`

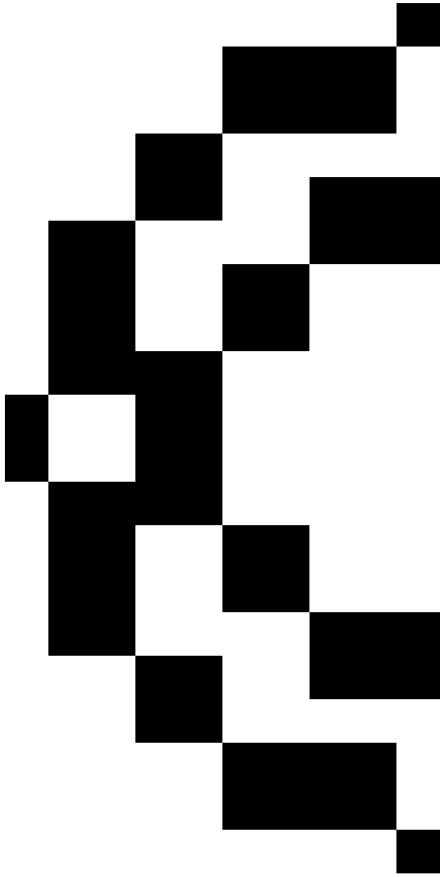


Exercise 9

Ask the user to insert 4 integers and store them in a list. Print `"high"` on the console, but only once, if the list contains at least one value over 600. It is enough for the program to find a single qualifying number to print the message.

Example

- `Input:  numbers = 200, 750, 300, 100`
- `Output: high`



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Thank you

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